

Differential Calculus

*Home work sheet # 12

Conic Section in Polar Coordinate system

7–8 Find the distances from the pole to the vertices, and then apply Formulas (8)–(10) to find the equation of the ellipse in rectangular coordinates. ■

7. (a) $r = \frac{6}{2 + \sin \theta}$ (b) $r = \frac{1}{2 - \cos \theta}$

8. (a) $r = \frac{6}{5 + 2 \cos \theta}$ (b) $r = \frac{8}{4 - 3 \sin \theta}$

9–10 Find the distances from the pole to the vertices, and then apply Formulas (12)–(14) to find the equation of the hyperbola in rectangular coordinates. ■

9. (a) $r = \frac{3}{1 + 2 \sin \theta}$ (b) $r = \frac{5}{2 - 3 \cos \theta}$

10. (a) $r = \frac{4}{1 - 2 \sin \theta}$ (b) $r = \frac{15}{2 + 8 \cos \theta}$

Cylindrical and Spherical Coordinate system

1–2 Convert from rectangular to cylindrical coordinates. ■

1. (a) $(4\sqrt{3}, 4, -4)$ (b) $(-5, 5, 6)$
(c) $(0, 2, 0)$ (d) $(4, -4\sqrt{3}, 6)$

2. (a) $(\sqrt{2}, -\sqrt{2}, 1)$ (b) $(0, 1, 1)$
(c) $(-4, 4, -7)$ (d) $(2, -2, -2)$

3–4 Convert from cylindrical to rectangular coordinates. ■

3. (a) $(4, \pi/6, 3)$ (b) $(8, 3\pi/4, -2)$
(c) $(5, 0, 4)$ (d) $(7, \pi, -9)$

4. (a) $(6, 5\pi/3, 7)$ (b) $(1, \pi/2, 0)$
(c) $(3, \pi/2, 5)$ (d) $(4, \pi/2, -1)$

5–6 Convert from rectangular to spherical coordinates. ■

5. (a) $(1, \sqrt{3}, -2)$ (b) $(1, -1, \sqrt{2})$
(c) $(0, 3\sqrt{3}, 3)$ (d) $(-5\sqrt{3}, 5, 0)$

6. (a) $(4, 4, 4\sqrt{6})$ (b) $(1, -\sqrt{3}, -2)$
(c) $(2, 0, 0)$ (d) $(\sqrt{3}, 1, 2\sqrt{3})$

7–8 Convert from spherical to rectangular coordinates. ■

7. (a) $(5, \pi/6, \pi/4)$ (b) $(7, 0, \pi/2)$
(c) $(1, \pi, 0)$ (d) $(2, 3\pi/2, \pi/2)$

8. (a) $(1, 2\pi/3, 3\pi/4)$ (b) $(3, 7\pi/4, 5\pi/6)$
(c) $(8, \pi/6, \pi/4)$ (d) $(4, \pi/2, \pi/3)$

*These problems are for the students only as home work. Search the reference books for more examples.